

Model Fairness Audit: UCI Adult Income - Random Forest

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1. Executive summary

51 /100
Fairness
score

Verdict: Fail

Primary metric: Equal opportunity. Are truly qualified people approved at similar rates?

Pass threshold: True-positive-rate ratio of at least 0.80.

Key findings

- 5 of 17 groups fail equal opportunity; the most affected is 'Female, Black' (value 0.605).
- 3 small group(s) fall below the threshold and need more data before a conclusion can be drawn.
- For context, the lowest selection-rate ratio is 0.133 (the four-fifths rule requires 0.80).
- Overall drift is stable (highest PSI 0.0079); 0 feature(s) shifted.
- Decision explanations reproduce the model's decisions on 97% of records.
- 'relationship' is strongly associated with 'sex' (Cramer's V 0.64) and may act as a proxy.

Priority actions

- High Investigate the equal opportunity gap for 'Female, Black' (sex x race): measured 0.605; required: true-positive-rate ratio of at least 0.80.
- High Investigate the equal opportunity gap for 'Female, White' (sex x race): measured 0.666; required: true-positive-rate ratio of at least 0.80.
- High Investigate the equal opportunity gap for 'Male, Black' (sex x race): measured 0.68; required: true-positive-rate ratio of at least 0.80.

2. Scope: data and model

Model	UCI Adult Income - Random Forest
Intended use	Lending
Records audited	9,049
Columns	16
Model decision column	predicted_high_income
Actual outcome column	income_over_50k
Sensitive attributes	sex, race
Overall selection rate	18%
Overall accuracy	86%
Positive outcome base rate	25%

3. Method and metric selection

Records are grouped by each sensitive attribute and by every combination of two or more attributes (intersectional groups). Each group is compared with a reference group: the best-off group with at least 100 records. The primary metric, **Equal opportunity**, decides pass or fail; the other metrics are reported for context.

Metric	Question answered	Pass threshold
Demographic parity	Are groups selected at similar rates?	Selection-rate ratio of at least 0.80
Equal opportunity	Are truly qualified people approved at similar rates?	TPR ratio of at least 0.80
Equalized odds	Are misses and false alarms similar across groups?	TPR and FPR gaps of at most 0.10
Predictive parity	Does a positive decision mean the same for every group?	Precision ratio of at least 0.80

Groups below 100 records are marked low confidence; if they fail they are marked "Review" rather than "Fail". Selection rates include 95% Wilson confidence intervals. The fairness score blends average and worst-group compliance with the primary metric, minus a penalty per failing group, and is capped below 70 whenever any group fails.

4. Fairness results: single attributes

sex

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
Male (reference)	6,144	23% (22%-24%)	1.00	59%	7%	1.00	0.00	0.99	1.00	Pass
Female	2,905	7% (6%-8%)	0.30	46%	2%	0.78	0.13	1.00	0.78	Fail

race

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
Asian-Pac-Islander (reference)	271	23% (18%-28%)	1.00	65%	7%	1.00	0.00	0.90	1.00	Pass
White	7,791	19% (18%-20%)	0.84	57%	5%	0.88	0.08	0.92	0.88	Pass
Amer-Indian-Eskimo	90	9% (5%-17%)	0.40	56%	4%	0.85	0.10	0.73	0.85	Pass
Other	57	9% (4%-19%)	0.39	60%	4%	0.92	0.05	0.70	0.92	Pass
Black	840	8% (6%-10%)	0.34	46%	1%	0.70	0.20	1.00	0.70	Fail

5. Fairness results: intersectional groups

sex x race

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
Male, Asian-Pac-Islander (reference)	189	29% (23%-35%)	1.00	69%	9%	1.00	0.00	0.91	1.00	Pass
Male, White	5,440	24% (23%-25%)	0.84	59%	7%	0.85	0.10	0.90	0.85	Pass
Male, Black	418	11% (9%-15%)	0.40	47%	2%	0.68	0.22	1.00	0.68	Fail
Male, Amer-Indian-Eskimo	58	10% (5%-21%)	0.36	60%	6%	0.86	0.09	0.57	0.86	Pass
Male, Other	39	10% (4%-24%)	0.36	50%	6%	0.72	0.19	0.57	0.72	Review
Female, Asian-Pac-Islander	82	9% (4%-17%)	0.30	40%	4%	0.58	0.29	0.65	0.58	Review
Female, White	2,351	7% (6%-8%)	0.26	46%	2%	0.67	0.23	0.92	0.67	Fail
Female, Amer-Indian-Eskimo	32	6% (2%-20%)	0.22	50%	0%	0.72	0.19	1.14	0.72	Review
Female, Other	18	6% (1%-26%)	0.20	100%	0%	1.44	0.31	1.14	1.44	Pass
Female, Black	422	4% (2%-6%)	0.13	42%	1%	0.60	0.27	0.93	0.60	Fail

6. Drift analysis

Population Stability Index (PSI): below 0.10 stable, 0.10 to 0.25 moderate, above 0.25 significant. The Kolmogorov-Smirnov p-value is given for numeric features (below 0.05 indicates a significant shift).

Feature	PSI	KS p-value	Band
native_country	0.008	n/a	Stable
education	0.003	n/a	Stable
age	0.002	0.1997	Stable
hours_per_week	0.002	0.8499	Stable
occupation	0.002	n/a	Stable
relationship	0.001	n/a	Stable
workclass	0.001	n/a	Stable
education_num	0.001	0.9563	Stable
age_band	0.001	n/a	Stable
marital_status	0.001	n/a	Stable
capital_gain	0.000	1.0	Stable
capital_loss	0.000	0.9801	Stable

7. Explainability

Explanations are produced by a transparent surrogate model that reproduces the audited model's decisions on **97%** of records. Sensitive attributes are excluded from explanations and counterfactuals.

Overall drivers of the model's decisions

Feature	Share of influence
relationship	19%
education	17%
marital status	16%
occupation	14%
capital gain	10%
age band	7%
education num	6%
hours per week	2%

Example decisions

Record #0 Rejected

The model rejected this record, mainly because of capital gain (0), compared with a typical record. Relationship (Husband) counted in its favour, but not enough to change the outcome.

Main factors: relationship = Husband (helped), marital status = Married-civ-spouse (helped), occupation = Prof-specialty (helped), capital gain = 0 (hurt)

What would change the outcome

- If capital gain were increased from 0 to 3,005, the decision would likely be approved.
- If education num were increased from 11 to 15, the decision would likely be approved.
- If education were 'Bachelors' instead of 'Assoc-voc', the decision would likely be approved.

Record #2 Approved

The model approved this record, mainly because of education (Bachelors) and relationship (Husband), compared with a typical record.

Main factors: education = Bachelors (helped), relationship = Husband (helped), marital status = Married-civ-spouse (helped), education num = 13 (helped)

What would change the outcome

- If education num were reduced from 13 to 8, the decision would likely be rejected.
- If education were 'HS-grad' instead of 'Bachelors', the decision would likely be rejected.
- If marital status were 'Never-married' instead of 'Married-civ-spouse', the decision would likely be rejected.

Possible proxy features

Feature	Sensitive attribute	Cramer's V
relationship	sex	0.64

8. Limitations

- Fairness metrics can conflict mathematically when groups have different base rates; passing the primary metric does not imply passing every metric.
- Results describe the supplied data only. If outcomes were recorded under a biased historical process, outcome-based metrics inherit that bias.
- Groups with fewer than 100 records are reported with low confidence and are not counted as firm failures.
- Per-decision explanations come from a transparent surrogate model that approximates the audited model; its agreement rate is stated in section 6.
- Counterfactuals change one feature at a time within the observed range and indicate sensitivity; they are not guaranteed routes to a different outcome.

- Proxy detection measures statistical association only and does not show that the model relies on the feature.
- This report supports, but does not replace, legal and domain review.

9. Recommendations

- **High** Investigate the equal opportunity gap for 'Female, Black' (sex x race): measured 0.605; required: true-positive-rate ratio of at least 0.80.
- **High** Investigate the equal opportunity gap for 'Female, White' (sex x race): measured 0.666; required: true-positive-rate ratio of at least 0.80.
- **High** Investigate the equal opportunity gap for 'Male, Black' (sex x race): measured 0.68; required: true-positive-rate ratio of at least 0.80.
- **Medium** 3 small group(s) fall below the threshold but have fewer than 100 people, for example 'Male, Other'. Validate with more data before drawing conclusions.
- **Medium** 'relationship' is strongly associated with 'sex' (Cramer's V 0.64). It may act as a proxy; review whether the model should use it.
- **Low** Schedule a fairness and drift re-check every quarter.

10. Audit trail and sign-off

Report generated	2026-10-07 12:14 UTC
Tool	FairLens 1.2 (Python 3.11.15)
Groups tested	17 (7 with fewer than 100 records)

Role	Name	Signature	Date
Model owner			
Independent reviewer			
Compliance / risk			